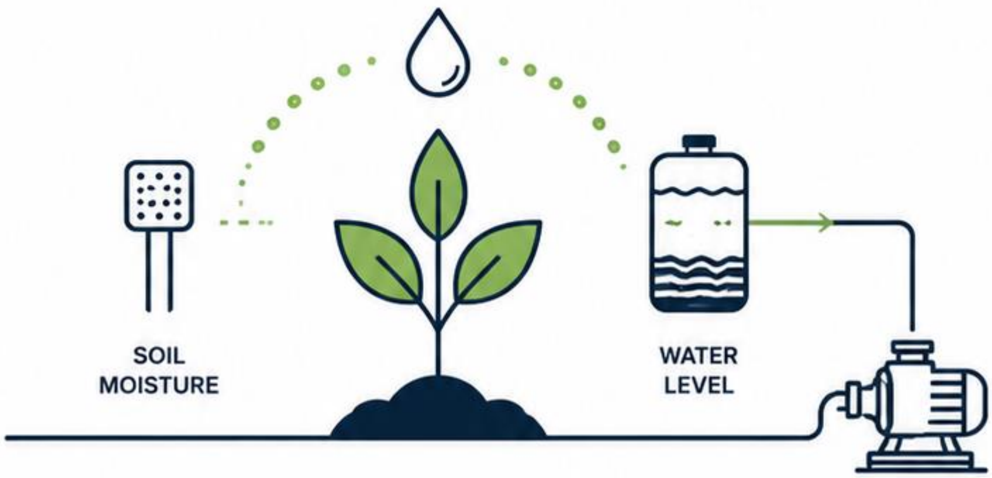
SMART
IRRIGATION SYSTEM

CUSTOMER REQUIREMENTS SPECIFICATION (CRS)

Embedded Real-Time Irrigation Control System



AUTOMATIC IRRIGATION | TANK PROTECTION | ERROR ALERTS | REAL-TIME MONITORING

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PROJECT FOCUS
Real-Time Irrigation Control • Sensor Monitoring
Safety Logic • Embedded System Design

PLATFORM
STM32F103C8

DOCUMENT VERSION
Version 1.0

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Designed as a self-directed embedded systems project focused on real-time control, reliability, safety, and professional engineering documentation.

1. Document Control

Item	Description
Document Title	Smart Irrigation System – Customer Requirements Specification
Document Type	CRS
Version	1.0
Project Platform	STM32F103C8
Development Environment	STM32CubeIDE, STM32CubeMX, Proteus
Purpose	Define the customer-facing requirements of the Smart Irrigation System.

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2. Introduction

This document defines the customer requirements for the Smart Irrigation System. The purpose of the system is to monitor soil moisture and control irrigation automatically while providing manual control, local monitoring, warning indicators, and protection against unsafe operating conditions.

The document describes what the customer expects the system to do. It does not describe internal software implementation, RTOS tasks, source code structure, queue names, or detailed hardware driver logic.

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3. Project Background

Traditional irrigation methods often depend on manual operation or fixed watering schedules. These methods may lead to water waste, over-watering, under-watering, or pump operation when water is not available.

The Smart Irrigation System is intended to improve irrigation control by monitoring soil moisture and automatically operating a pump only when watering is required. The system also provides a manual watering option, tank-level safety protection, visible status feedback, and error indication.

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4. Project Scope

The Smart Irrigation System shall provide the following high-level capabilities:

- Monitor soil moisture continuously.
- Determine whether soil is wet or dry.
- Start watering automatically when soil is dry.
- Stop watering when soil becomes wet.
- Support timed manual watering.
- Prevent pump operation when the water tank is empty.
- Detect selected system faults.
- Provide status information through OLED and UART.
- Provide visual and audible error indication.
- Support reset-based recovery from selected error conditions.

The project scope is limited to a single irrigation zone with one soil moisture sensor, one water tank level sensor, one pump output, one OLED display, and one UART monitoring interface.

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5. Customer Requirements

The customer requires a smart irrigation controller that can make watering decisions based on soil conditions and protect the pump from unsafe operation.

The system shall:

1. Automatically water dry soil.
2. Stop watering when soil becomes wet.
3. Allow the user to request manual watering.
4. Prevent the pump from operating when the water tank is empty.
5. Notify the user when a fault occurs.
6. Display current system information locally and through UART.
7. Limit repeated watering attempts if soil remains dry.
8. Allow controlled recovery through a reset button.
9. Continue indicating that the controller is alive during normal operation.
10. Operate reliably in a Proteus simulation environment.

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6. Functional Requirements

ID	Functional Requirement
FR-01	The system shall read soil moisture using an analog sensor input.
FR-02	The system shall classify soil condition as WET or DRY.
FR-03	The system shall calculate and display a soil moisture percentage.
FR-04	The system shall detect dry soil and initiate automatic watering when the tank is available.
FR-05	The system shall detect wet soil and stop automatic watering.
FR-06	The system shall turn the pump ON during automatic watering.
FR-07	The system shall turn the pump OFF when watering is complete or soil becomes wet.
FR-08	The system shall wait for a defined rest period between automatic watering attempts.
FR-09	The system shall count automatic watering attempts.
FR-10	The system shall limit the maximum number of automatic watering attempts.
FR-11	The system shall enter an error condition if soil remains dry after the maximum allowed attempts.
FR-12	The system shall support manual watering through a push button.
FR-13	The system shall limit manual watering to a predefined time period.
FR-14	The system shall block manual watering when the water tank is empty.
FR-15	The system shall monitor the water tank level.
FR-16	The system shall immediately stop the pump when the water tank becomes empty.
FR-17	The system shall report a water tank error when the tank is empty.
FR-18	The system shall keep the water tank error active until the tank is available and the user presses reset.
FR-19	The system shall allow selected recoverable errors to be cleared by a reset button.
FR-20	The system shall detect invalid soil sensor readings.
FR-21	The system shall select and report the highest-priority active error.
FR-22	The system shall activate an error LED when an error is active.
FR-23	The system shall activate a buzzer when an error is active.
FR-24	The system shall toggle a heartbeat LED during normal operation.
FR-25	The system shall display current system status on an OLED display.
FR-26	The system shall send system status information through UART.
FR-27	The system shall report pump status, soil status, tank status, current error, and watering attempts.
FR-28	The system shall prevent unsafe pump operation during critical fault conditions.

ID	Functional Requirement
FR-29	The system shall provide user-visible confirmation of system recovery after reset.
FR-30	The system shall operate as one complete irrigation control unit in Proteus simulation.

7. Non-Functional Requirements

ID	Non-Functional Requirement
NFR-01	The system shall provide a response to a water tank empty condition without unsafe pump delay.
NFR-02	The pump control logic shall have higher importance than reporting and display functions.
NFR-03	The system shall provide readable OLED status information.
NFR-04	The system shall provide readable UART status reports.
NFR-05	The system shall avoid rapid switching between WET and DRY states near threshold boundaries.
NFR-06	The system shall prevent multiple false button actions caused by button bouncing.
NFR-07	The system shall keep pump control separate from user interface reporting.
NFR-08	The system shall provide an error indication through more than one method.
NFR-09	The system shall be modular and suitable for future expansion.
NFR-10	The system shall protect shared communication resources.
NFR-11	The system shall maintain safe pump behavior during error conditions.

8. Safety Requirements

ID	Safety Requirement
SR-01	The pump shall be forced OFF when the water tank is empty.
SR-02	Manual watering shall not override water tank protection.
SR-03	The pump shall not continue operating indefinitely if soil remains dry.
SR-04	A water tank error shall require user reset before normal recovery.
SR-05	Critical errors shall be clearly visible through LED, buzzer, OLED, and UART feedback.
SR-06	The system shall prioritize water tank error above lower-priority errors.
SR-07	The system shall stop the pump if soil becomes wet during watering.
SR-08	The system shall detect invalid ADC readings and notify the user.

9. Customer and Project Constraints

The following constraints define the boundaries of the current Smart Irrigation System project. These constraints are based on the agreed project scope and validation environment rather than internal implementation choices.

Constraint ID	Constraint
CPC-01	The system shall control one irrigation zone only in the current project version.
CPC-02	The system shall monitor one soil area using one soil moisture measurement point.
CPC-03	The system shall control one pump or irrigation relay output.
CPC-04	The system shall provide local system status feedback to the user.
CPC-05	The system shall provide a method for manual watering and error reset.
CPC-06	The system shall provide protection against pump operation when water is unavailable.
CPC-07	Real field deployment, waterproof enclosure design, and long-term environmental testing are outside the scope of the current version.
CPC-08	Cloud connectivity, mobile application control, and multi-zone irrigation are outside the scope of the current version.
CPC-09	Soil moisture thresholds may require calibration before real hardware deployment.

10. Acceptance Criteria

The project shall be accepted when all of the following conditions are satisfied:

Acceptance ID	Acceptance Criterion
AC-01	The pump remains OFF when soil is wet.
AC-02	The pump turns ON automatically when soil is dry and tank is available.
AC-03	The pump turns OFF when soil becomes wet during watering.
AC-04	Manual watering operates only when the tank is available.
AC-05	Manual watering is blocked when the tank is empty.
AC-06	The pump turns OFF immediately when the tank becomes empty.
AC-07	Water tank error remains active until reset is pressed after tank recovery.
AC-08	The system enters an error state after maximum watering attempts are reached.
AC-09	Invalid ADC readings generate an ADC sensor error indication.
AC-10	The highest-priority active error is displayed to the user.
AC-11	OLED display and UART terminal show valid system status information.
AC-12	Error LED and buzzer operate while an error is active.
AC-13	Heartbeat LED continues toggling during normal system operation.

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11. User Interaction Requirements

User Action	Expected System Response
User presses Manual Watering button	System starts timed manual watering only when tank is OK and no critical blocking error is active.
User presses Reset button during a recoverable error	System clears the error only when the related recovery condition is valid.
User observes OLED	OLED displays state, moisture, pump, tank, attempts, and error information.
User observes UART terminal	UART displays periodic system status reports and important events.
User observes Error LED and buzzer	Both indicate that an error is active.
User observes Heartbeat LED	LED toggling confirms that the system is operating.

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12. Risk Summary

Risk	Customer Impact	Required Protection
Empty water tank	Pump may run dry and be damaged	Pump shutdown and tank error latch.
Soil remains dry after irrigation	Water waste or pump stress	Maximum watering attempts protection.
User requests watering with empty tank	Unsafe manual pump operation	Manual watering blocking logic.
Soil sensor fault	Incorrect watering decision	ADC range validation and error reporting.
Multiple simultaneous errors	User may see wrong fault information	Priority-based error reporting.
Communication output conflict	Incorrect monitoring information	Protected UART and I2C communication.

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13. Requirements Traceability Matrix

Customer Need	Related Requirement IDs	Acceptance Criteria	Validation Test
Avoid watering wet soil	FR-02, FR-05, FR-07	AC-01, AC-03	BV-01, BV-03
Water dry soil automatically	FR-04, FR-06	AC-02	BV-02
Support manual watering	FR-12, FR-13	AC-04	BV-08
Protect pump from empty tank	FR-15, FR-16, FR-17, FR-18	AC-05, AC-06, AC-07	BV-06, BV-07, BV-09
Prevent endless watering	FR-08, FR-09, FR-10, FR-11	AC-08	BV-04, BV-05
Detect sensor problems	FR-20	AC-09	BV-10
Display critical errors correctly	FR-21, FR-22, FR-23	AC-10, AC-12	BV-11
Provide local and serial monitoring	FR-25, FR-26, FR-27	AC-11	All BV tests
Confirm system operation	FR-24	AC-13	BV-01
Validate full system behavior	FR-30	AC-14	BV-01 to BV-11

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14. Conclusion

This Customer Requirements Specification defines the expected behavior of the Smart Irrigation System from the customer perspective. The system is required to monitor soil moisture, control irrigation automatically and manually, protect the pump from unsafe conditions, provide error notification, and support user monitoring through OLED, UART, LEDs, and buzzer outputs.

The requirements defined in this document form the baseline for the Software Requirements Specification, System Design Document, implementation, and final validation testing.